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**NLP Practical 8**

**Movie Sentiment Analyzer**

**Github : https://github.com/PushpakBajanghate/Movie_Sentiment_Analyzer**

Aim : Write the python code to perform sentiment analysis using NLP Instructions for Preparing the Movie Review Sentiment Analysis PDF Document.

DataSet Used : IMDB DataSet.csv

Code :

```
import pandas as pd

df = pd.read_csv("IMDB Dataset.csv", engine='python', on_bad_lines='skip')

import pandas as pd

import numpy as np

import nltk

import re

import matplotlib.pyplot as plt

import seaborn as sns

import pickle


from nltk.corpus import stopwords

from nltk.tokenize import word_tokenize


from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.model_selection import train_test_split
```

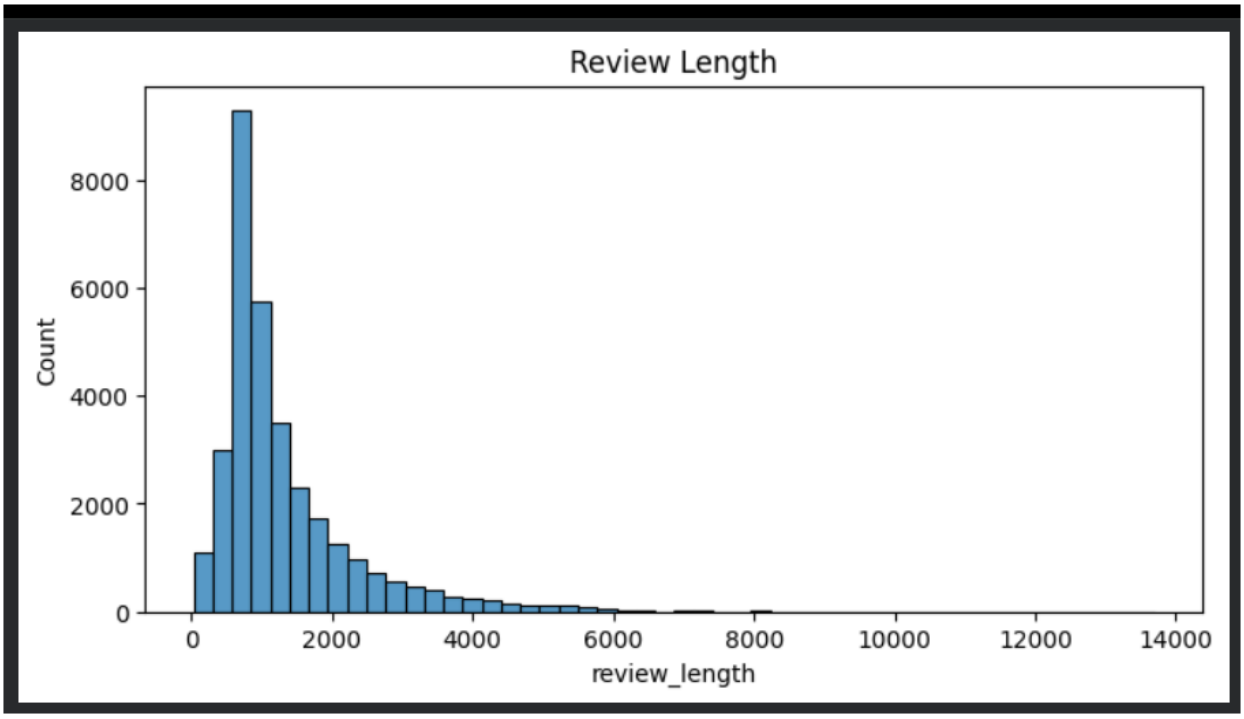
```
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score,
precision_score, recall_score, f1_score


from sklearn.naive_bayes import MultinomialNB

from sklearn.linear_model import LogisticRegression

from sklearn.svm import LinearSVC

from sklearn.ensemble import RandomForestClassifier


from wordcloud import WordCloud


import spacy

import gradio as gr

nltk.download('punkt')

nltk.download('stopwords')

nltk.download('punkt_tab')

df = pd.read_csv("/content/IMDB Dataset.csv", engine='python', on_bad_lines='skip')


df.head()
```

| | review | sentiment |
|---|---|---|
| 0 | One of the other reviewers has mentioned that ... | positive |
| 1 | A wonderful little production. <br /><br />The... | positive |
| 2 | I thought this was a wonderful way to spend ti... | positive |
| 3 | Basically there's a family where a little boy ... | negative |
| 4 | Petter Mattei's "Love in the Time of Money" is... | positive |

Next steps: Generate code with df  New interactive sheet

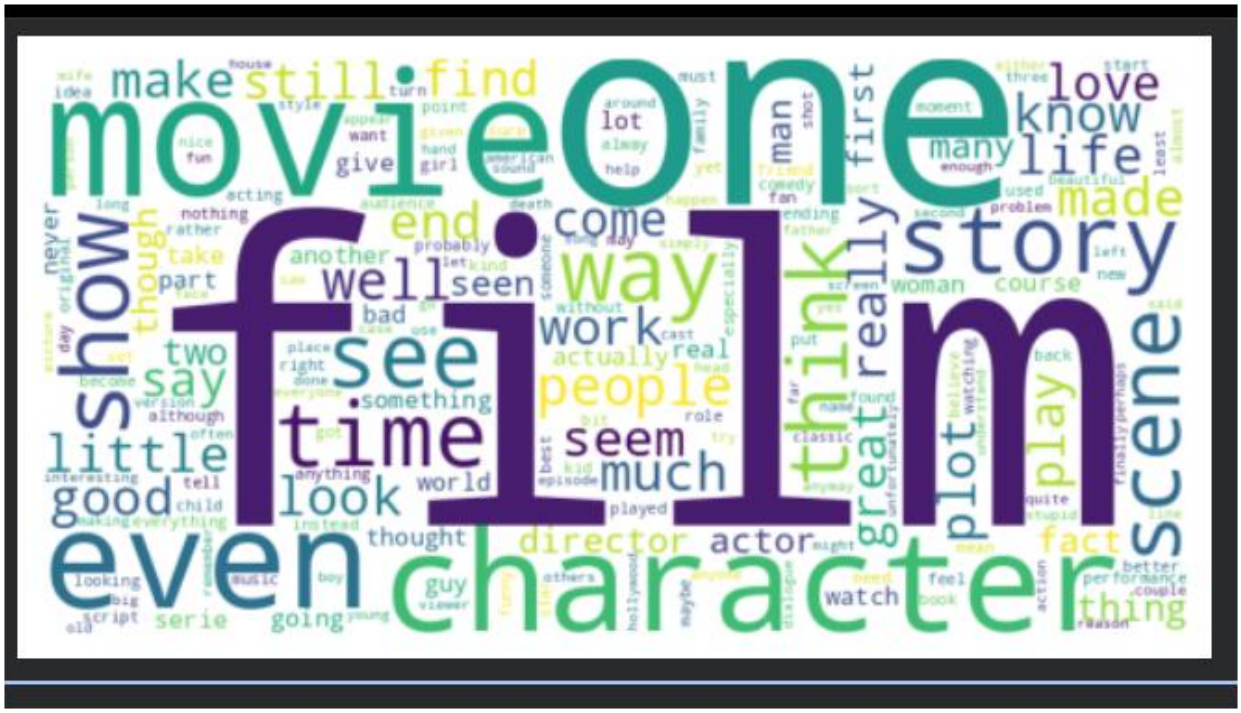
```
print(df.columns)


text_col = df.columns[0]

target_col = df.columns[1]


df = df[[text_col, target_col]]

df.columns = ["review", "sentiment"]

df.shape

df.info()

sns.countplot(x="sentiment", data=df)

plt.title("Sentiment Distribution")

plt.show()

df["review_length"] = df["review"].apply(len)
```

```
plt.figure(figsize=(8,4))

sns.histplot(df["review_length"], bins=50)

plt.title("Review Length")

plt.show()
```




```
print(df.isnull().sum())


print("Duplicates:", df.duplicated().sum())


df.dropna(inplace=True)

df.drop_duplicates(inplace=True)

from nltk.stem import WordNetLemmatizer

nltk.download('wordnet')


lemmatizer = WordNetLemmatizer()
```

```
stop_words = set(stopwords.words("english"))

def preprocess(text):

    text = str(text).lower()
    text = re.sub(r"<.*?>"," ",text)
    text = re.sub(r"http\S+"," ",text)
    text = re.sub(r"[^a-zA-Z]"," ",text)

    tokens = word_tokenize(text)
    tokens = [w for w in tokens if w not in stop_words and len(w)>2]

    tokens = [lemmatizer.lemmatize(w) for w in tokens]

    return " ".join(tokens)

df["clean_text"] = df["review"].apply(preprocess)
text = " ".join(df["clean_text"])

wordcloud = WordCloud(width=800,height=400,background_color="white").generate(text)

plt.imshow(wordcloud)
plt.axis("off")
plt.show()
```


film
one
movie
character
even
time
story
scene
see
way
people
little
show
good
look
work
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much
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child
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music
boy
script
old
interesting
making everything
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episode
best
kid
hollywood
maybe
others
feel
book
action
reason
better
performance
couple
stupid
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finally perhaps
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watching
understand
believe
classic
unfortunately
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put
screen
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second
problem
beautiful
used
enough
fun
nice
house
wife
idea
turn
point
style
appear
hand
girl
sound
audience
death
fan
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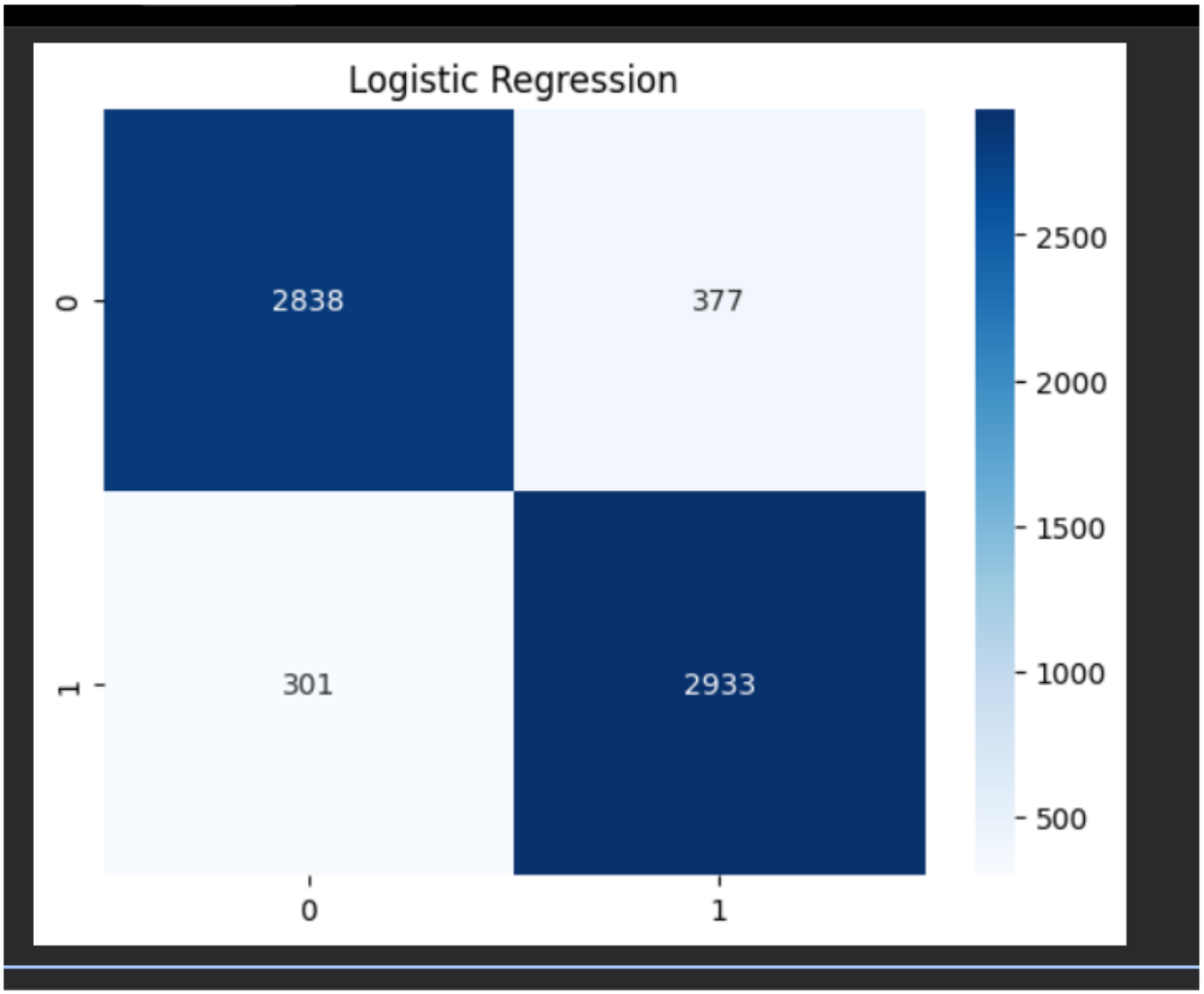
```
max_df=0.9
)

X = vectorizer.fit_transform(df["clean_text"])
y = df["sentiment"]
X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42, stratify=y
)

vectorizer = TfidfVectorizer(
    max_features=25000,
    ngram_range=(1,2),
    min_df=2,
    max_df=0.9
)

X = vectorizer.fit_transform(df["clean_text"])
y = df["sentiment"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
models = {
"Naive Bayes": MultinomialNB(),
"Logistic Regression": LogisticRegression(max_iter=2000),
```

```
"Linear SVC": LinearSVC(C=1.8),
"Random Forest": RandomForestClassifier(n_estimators=400)
}
results = []

for name, model in models.items():

    model.fit(X_train, y_train)
    pred = model.predict(X_test)

    results.append([
        name,
        accuracy_score(y_test, pred),
        precision_score(y_test, pred),
        recall_score(y_test, pred),
        f1_score(y_test, pred)
    ])

results_df = pd.DataFrame(
results,
columns=["Model","Accuracy","Precision","Recall","F1 Score"]
)

results_df
```

| | Model | Accuracy | Precision | Recall | F1 Score |
|---|---|---|---|---|---|
| 0 | Naive Bayes | 0.880609 | 0.872960 | 0.891903 | 0.882330 |
| 1 | Logistic Regression | 0.900272 | 0.890674 | 0.913402 | 0.901895 |
| 2 | Linear SVC | 0.892004 | 0.890132 | 0.895318 | 0.892718 |
| 3 | Random Forest | 0.864273 | 0.859434 | 0.872212 | 0.865776 |

Next steps: Generate code with results_df New interactive sheet

```
best_model_name = results_df.loc[results_df["Accuracy"].idxmax(),"Model"]

best_model = models[best_model_name]

best_model.fit(X_train, y_train)

pred = best_model.predict(X_test)

cm = confusion_matrix(y_test, pred)

sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.title(best_model_name)
plt.show()
```

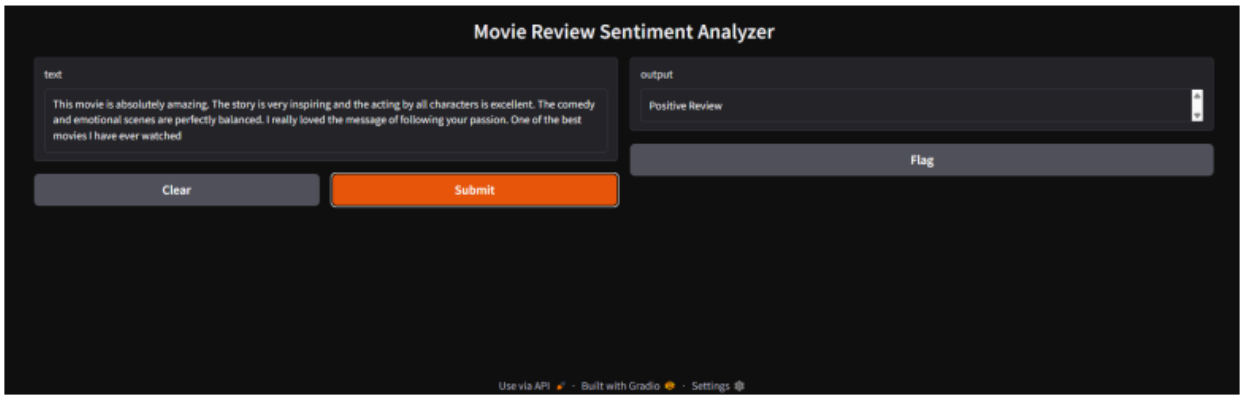


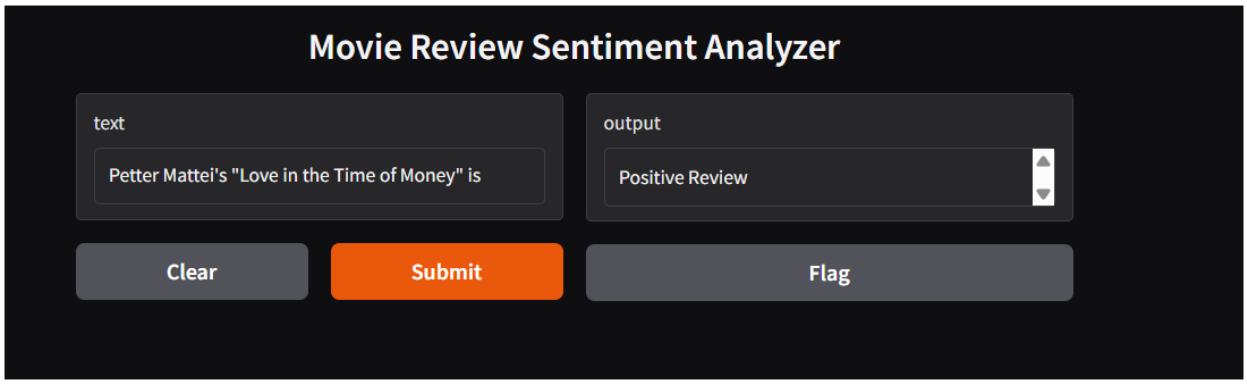

pickle.dump(best_model, open("sentiment_model.pkl","wb")) pickle.dump(vectorizer, open("vectorizer.pkl","wb"))

def predict(text): clean = preprocess(text) vec = vectorizer.transform([clean]) return best_model.predict(vec)[0] predict("This movie is absolutely amazing")

def predict_sentiment(text): result = predict(text) return "Positive Review" if result==1 else "Negative Review" interface = gr.Interface( fn=predict_sentiment, inputs="text", outputs="text", title="Movie Review Sentiment Analyzer" ) interface.launch()

Output :




Best Performance Model : Logistic Regression

Accuracy: 90.02%